Data Collection Technique and Tools

Introduction

- Data collection methods are scientific in procedure calling for different skills.
- The choice of data collection method(s) is determined by context and minimum cost enough for the purpose of the study
- Primary and Secondary data require different methods of data collection
- These methods can be grouped under Quantitative (numerical) and Qualitative (non-numerical)

DATA COLLECTION

- Defined as:
  - Compilation and interpretation of primary and secondary sources of information.
  - The integration of different sources will consolidate the write up of the report.

Types of Data Collection Methods

•Qualitative techniques are better suited to examine the feelings, beliefs, attitudes, or motivation that underlie an observed problem.

•They are helpful in exploring the causes of a problem, constraints to changes in behavior, and opportunities for correcting the problem.

•Qualitative methods are based on talking to people at length and in depth or observing their behavior.

Types of Data Collection Methods (cont'd)

•Quantitative techniques are used to describe patterns or to pinpoint specific problems that need attention, but they are usually not good for understanding why these patterns or problems exist.

SOURCES OF INFORMATION

Primary Source

•Data is collected by researcher himself

•Data is gathered through questionnaire, interviews, observations etc.

Secondary Source

•Data collected, compiled or written by other researchers eg. books, journals, newspapers

•Any reference must be acknowledged

METHODS USED TO COLLECT PRIMARY SOURCE DATA

1. Interviews
2. Questionnaires
3. Survey
4. Experimentation
5. Case Study
6. Observation

However, for a small-scale study, the most commonly used methods are interviews, survey questionnaires and observations.

Primary Data - Limitations

- Do you have the time and money for:
  - Designing your collection instrument?
  - Selecting your population or sample?
  - Pretesting/piloting the instrument to work out sources of bias?
  - Administration of the instrument?
  - Entry/collation of data?

Primary Data - Limitations

- Uniqueness
  - May not be able to compare to other populations
- Researcher error
  - Sample bias
  - Other confounding factors

Interview Method (Survey)

The interview method of collecting data involves presentation of oral, verbal stimuli and reply in terms of oral-verbal responses. This method can also be used through personal telephone interviews and self-administered questionnaire.

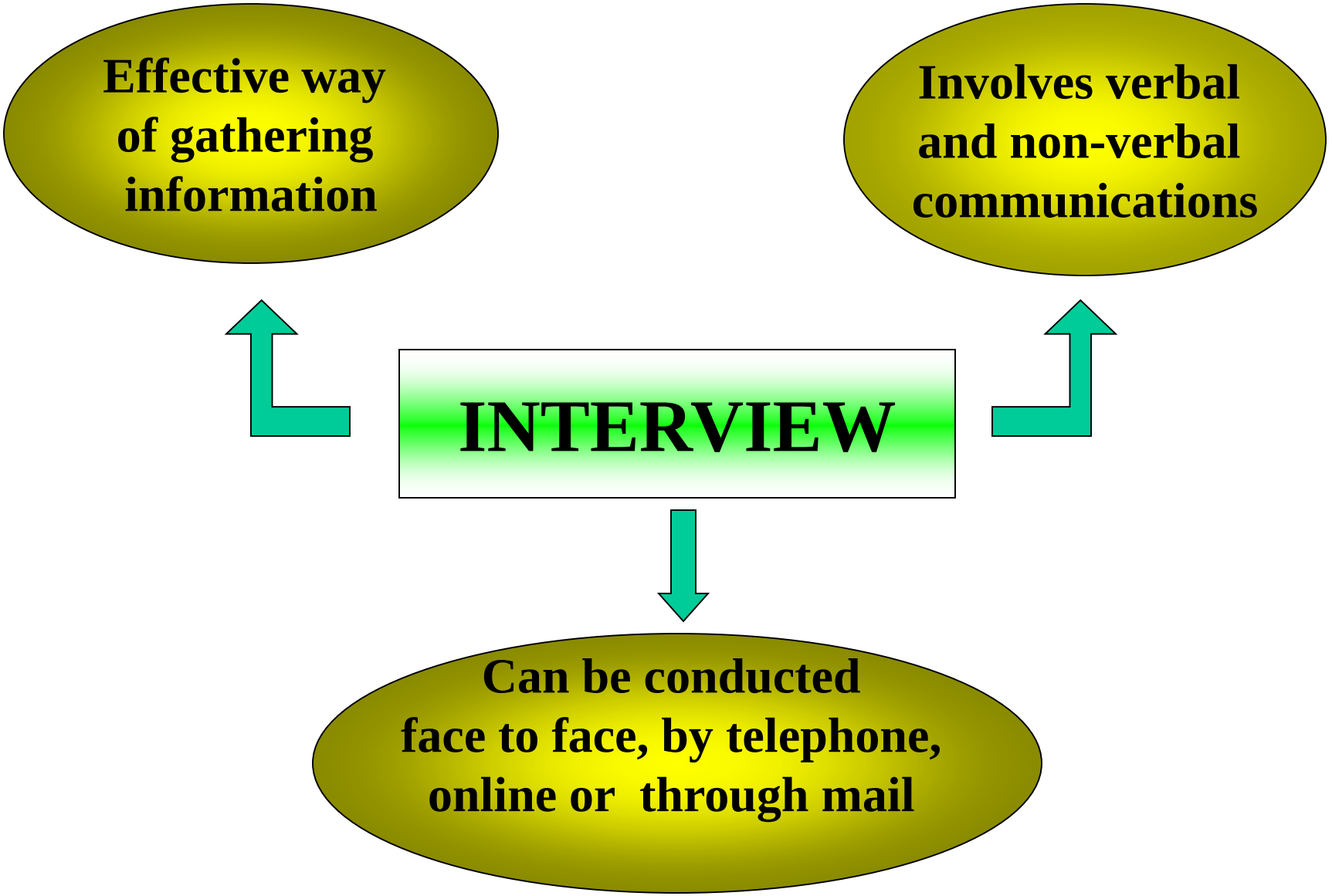
Personal interview: Personal interview method requires a person known as the interviewer asking questions generally in a face to face contacts to the other person or persons – respondent (s).

Interview Method (Cont'd)

In an interview method – Usually uses a fixed set of questions asked from sample of respondents who have been selected to represent a larger population. Such surveys can focus on facts about respondents or on what they know, or they can focus more on respondent's attitudes, opinions, and other less easily qualified dimensions.

Types of Interview

1. Structured or directive interview
2. Unstructured or non-directive interview
3. Focused interview
4. Clinical interview
5. Depth interview


Effective way of gathering information
Involves verbal and non-verbal communications
INTERVIEW
Can be conducted face to face, by telephone, online or through mail

Steps To An Effective Interview

Prepare your interview schedule



Select your subjects/ key informants



Conduct the interview



Analyze and interpret data collected from the interview

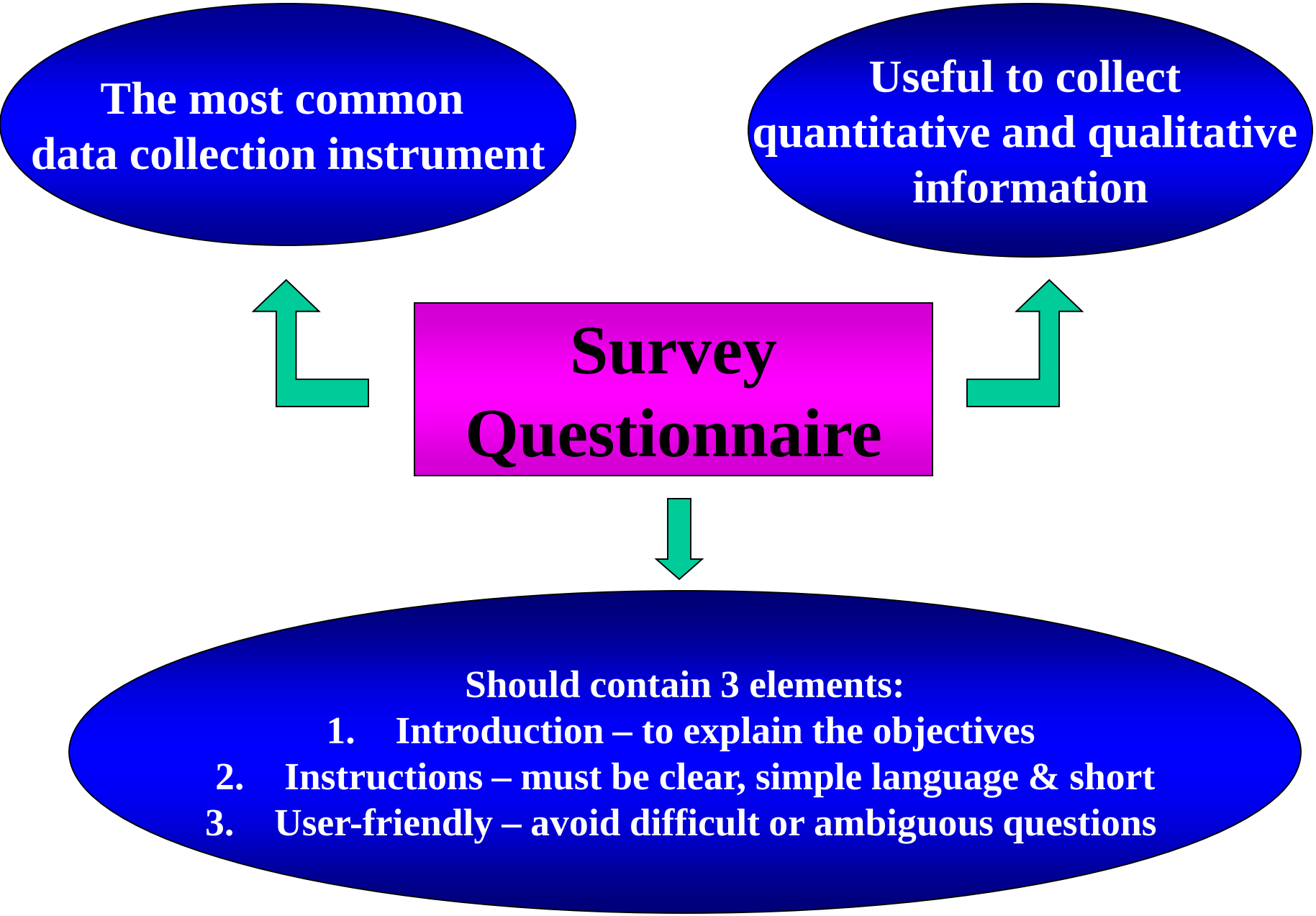
What is a focus group?

•A focus group is where people from similar backgrounds or experiences that get together to discuss specific topics of interest to the researcher.

•The group participants are guided by the moderator or facilitator, who introduces topics for discussion and helps the group participate in a lively and natural manner amongst themselves.

Advantages of Interviews

1. Response rates. Response rates tend to be high in face-to-face interviews.
2. Audience. Many people simply cannot fill out a questionnaire.
3. Clarity. Interviews offer some protection against ambiguous or confusing questions.
4. Depth of questioning. The information obtained from questionnaires tends to be more superficial than interview data,
5. Missing information. Respondents are less likely to give “don’t know” responses or to leave a question


The most common data collection instrument
Useful to collect quantitative and qualitative information
Survey Questionnaire
Should contain 3 elements:
1. Introduction – to explain the objectives
2. Instructions – must be clear, simple language & short
3. User-friendly – avoid difficult or ambiguous questions

Steps in Questionnaire Construction

1. Preparation
2. Constructing the first draft
3. Self-evaluation
4. External evaluation
5. Revision
6. Pre-test or Pilot study
7. Revision
8. Second Pre-test if necessary
9. Preparing final Copy

Questionnaire Design

Open-Ended Question	An interview question that encourages an answer phrased in respondent's own words.
Closed-Ended Question	An interview question that asks the respondent to make a selection from a limited list of responses.
Scaled-Response Question	A closed-ended question designed to measure the intensity of a respondent's answer.

Cont…

- Open-ended Questions

1. Free-response
2. (Text Open End)
3. Fill-in relevant information

- Close-ended Questions

1. Dichotomous question
2. Multiple-choice
3. Rank
4. Scale
5. Categorical
6. Numerical

Steps To An Effective Survey Questionnaire

Prepare your survey questions

(Formulate & choose types of questions, order them, write instructions, make copies)

⇩

Select your respondents/sampling

Random/Selected

⇩

Administer the survey questionnaire

(date, venue, time)

⇩

Tabulate data collected

(Statistical analysis-frequency/mean/correlation/%)

⇩

Analyze and interpret data collected

A sample of complete survey questionnaire

Advantages of Questionnaires

The strengths of questionnaires include the following:

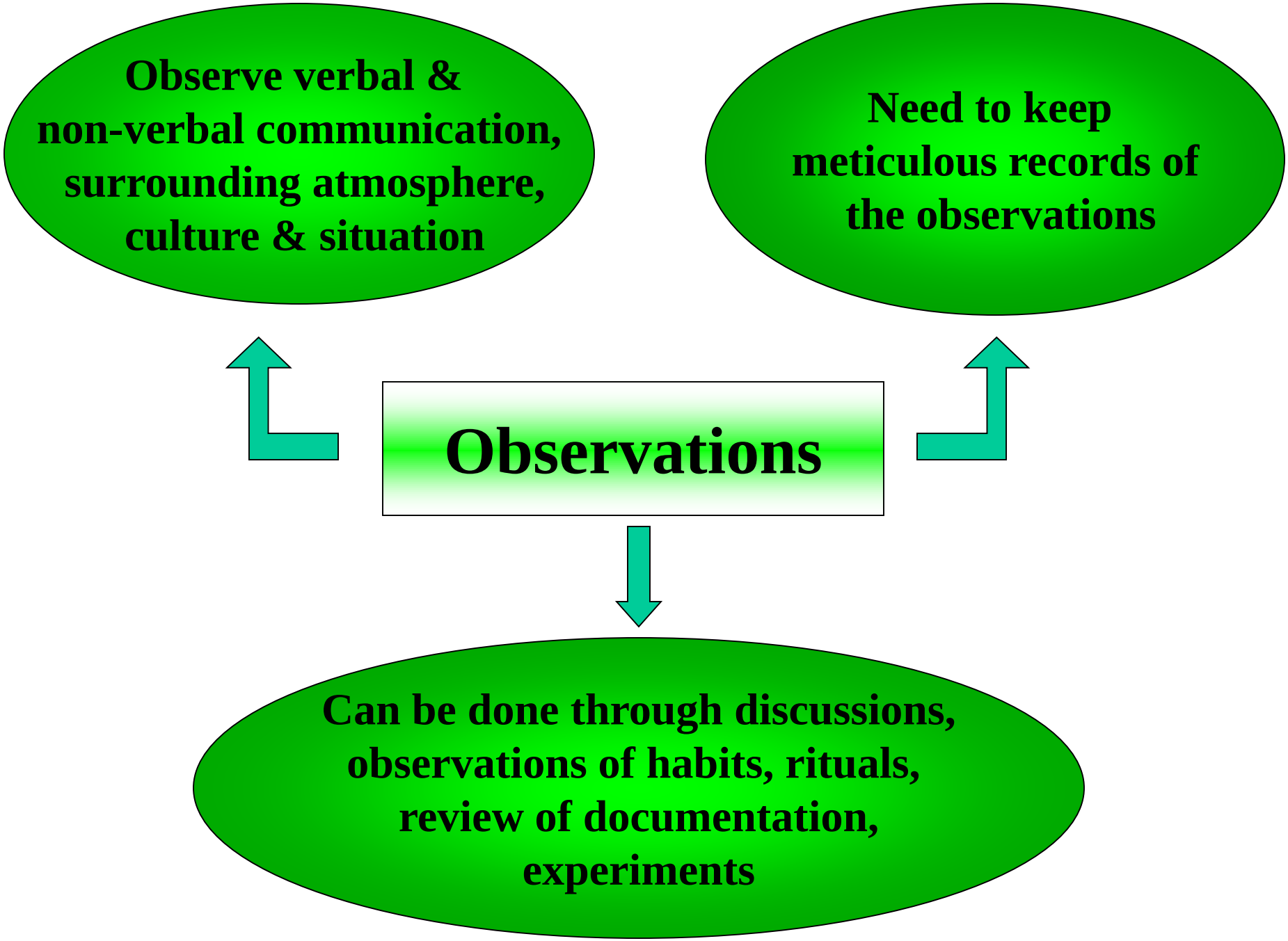
• Cost. Questionnaires, relative to interviews, are in general much less costly and require less time and energy to administer.

• Anonymity. Unlike interviews, questionnaires offer the possibility of complete anonymity.

Observation

- The observation method is the most commonly used method especially in studies relating to behavioral sciences. In a way we all observe things around us, though usually in unscientific manner.

- Observation becomes a scientific tool and the method of data collection for the researcher, when it serves a formulated research purpose, is systematically planned and recorded and is subjected to checks and control on validity and reliability.


Observe verbal & non-verbal communication, surrounding atmosphere, culture & situation
Need to keep meticulous records of the observations
Observations
Can be done through discussions, observations of habits, rituals, review of documentation, experiments

Steps To An Effective Observation

Determine what needs to be observed

(Plan, prepare checklist, how to record data)

⇩

Select your participants

Random/Selected

⇩

Conduct the observation

(venue, duration, recording materials, take photographs)

⇩

Compile data collected

⇩

Analyze and interpret data collected

TYPES OF TOOLS

•The various methods of data gathering involve the use of appropriate recording forms. These are called tools or instruments of data collection. They consist of :

•Observation schedule

•Interview guide

•Interview schedule

•Mailed questionnaire

- Rating scale
- Checklist
- Document schedule/data sheet
- Schedule for institutions

Selection of an appropriate data collection method

- ***Nature, scope and object of enquiry*:** This constitutes the most important factor affecting the choice of a particular method. The method selected should be such that it suits the type of enquiry that is to be conducted by the researcher. This factor is also important in deciding use of secondary or primary data or both.

Selection of an appropriate data collection method (cont…)

- ***Availability of Funds*:** Finance, in fact, is a big constraint in practice and the researcher has to act within this limitation. **V**ery limited fund may compel the researcher to select a comparatively cheaper method, which may not be as efficient and effective as some other costly method.

- ***Time Factor*:** The available time affects the selection of the method by which the data is to be collected. Some methods take relatively more time, while with others the data can be collected in a comparatively shorter time.

- ***Precision required:*** Precision required is yet another important factor to be considered at the time of selecting the method of collection of data.